Algebra 1
Final Exam

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MATH

Algebra 1 Final Exam

This exam is comprehensive over the entire course and includes 12 questions. You have 60 minutes to complete the exam.

The exam is worth 100 points. The 8 multiple choice questions are worth 5 points each (40 points total) and the 4 free response questions are worth 15 points each (60 points total).

Mark your multiple choice answers on this cover page. For the free response questions, show your work and make sure to circle your final answer.

1. (5 pts)	A	B	C	D	E
2. (5 pts)	A	B	C	D	E
3. (5 pts)	A	B	C	D	E
4. (5 pts)	A	B	C	D	E
5. (5 pts)	A	B	C	D	E
6. (5 pts)	A	B	C	D	E
7. (5 pts)	A	B	C	D	E
8. (5 pts)	A	B	C	D	E

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1. **(5 pts)** Use the distributive property to expand the expression.

$$\frac{2x}{yz}\left(\frac{4x}{3y}-\frac{x^2}{z}\right)$$

A $\frac{16x^2}{3yz}-\frac{2x^2}{yz}$

B $\frac{2x^2}{3y^2z}-\frac{2x^3}{yz^2}$

C $\frac{8x^2}{3y^2z}+\frac{2x^3}{yz^2}$

D $\frac{8x^2}{3y^3z}-\frac{2x^3}{yz^3}$

E $\frac{8x^2}{3y^2z}-\frac{2x^3}{yz^2}$

2. **(5 pts)** Which property is illustrated by $3 \cdot (4 \cdot 2) = 3 \cdot (2 \cdot 4)$?

A Commutative property

B Associative property

C Identity property

D Transitive property

E Distributive property

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3. **(5 pts)** Translate "three less than the quotient of six and a number" to a math expression.

A $3-\frac{6}{x}$

C $\frac{6}{x}-3$

E $\frac{6}{x}+3$

B $3-6x$

D $6x-3$

4. **(5 pts)** Solve $-(4x-10)+14=2(4x-6)$.

A -1

C $\frac{3}{4}$

E 3

B $-\frac{1}{4}$

D 2

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5. **(5 pts)** What is the domain of the function $f(x) = 3\sqrt{x}$?

A $x \neq 3$

B $x \geq 0$

C $x \neq 0$

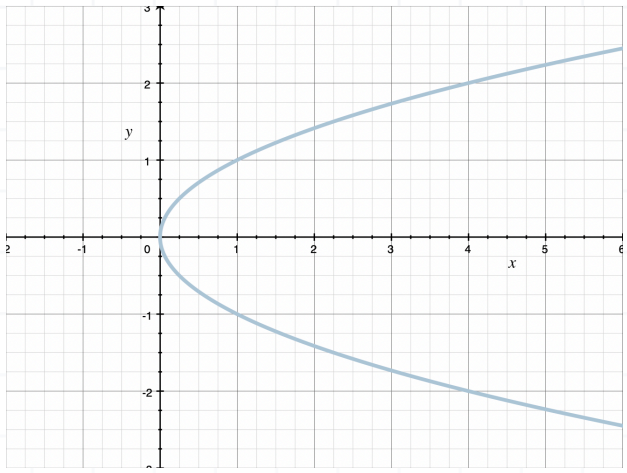
D $x \neq -3$

E $x \leq 0$

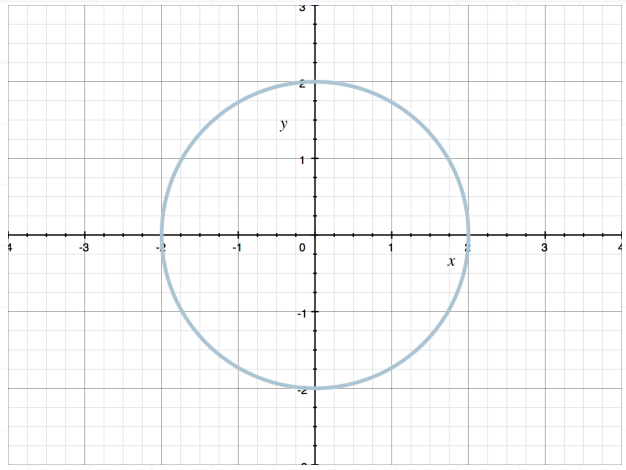
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6. **(5 pts)** Which graph represents a function?

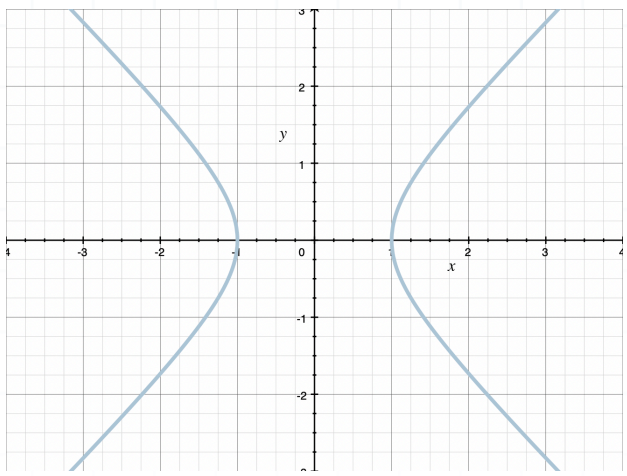
A



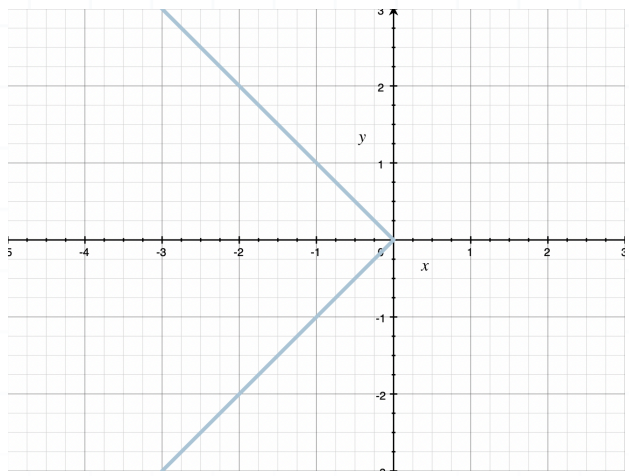
D



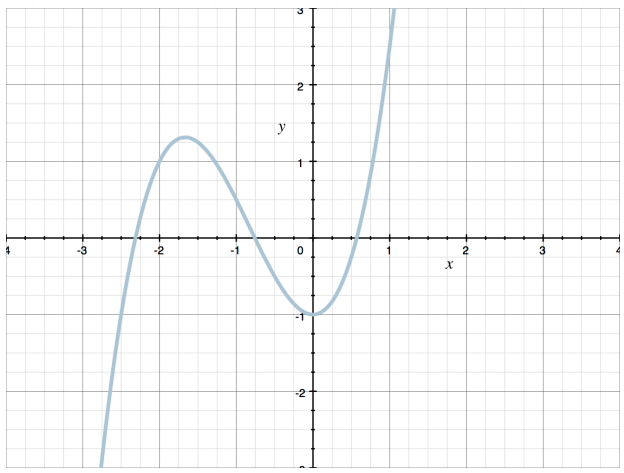
B



E



C



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7. **(5 pts)** Solve $5 - 4x \leq 13$.

A	$x \geq 8$	C	$x \leq -8$	E	$x \geq 2$
B	$x \leq 2$	D	$x \geq -2$		

8. **(5 pts)** Find $(fg)(-3)$ if $f(x) = x - 3$ and $g(x) = 2x + 5$.

A	-15	C	-3	E	7
B	-6	D	6		

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9. **(15 pts)** Simplify the expression by combining the two fractions.

$$\frac{3}{x+5}+\frac{6}{x^2+2x-15}$$

10. **(15 pts)** Solve the system of equations.

$$4x+6y=15$$

$$x=6-6y$$

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11. **(15 pts)** Find the solutions of $x^2 + 14x - 32 = 0$ by factoring.

12. **(15 pts)** Graph the linear inequality $-x - y > x - 5$.

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